

Linear Programming Course Summary

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1 Key Terms

- **Slack Variable:** A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. The slack variable represents the difference between the left-hand side and the right-hand side of the inequality.
- **Feasible Region:** The set of all points that satisfy all the constraints of a linear programming problem.
- **Vertex:** A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.
- **Basic Variable:** In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. Basic variables are set to values that satisfy the constraints.
- **Non-basic Variable:** In the Simplex method, a variable that is set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.
- **Pivot Operation:** The process of exchanging a basic variable with a non-basic variable in the Simplex method, leading to a new dictionary and a movement to a different vertex in the feasible region.
- **Dictionary:** A representation of a linear programming problem in a tabular format, where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move from one vertex to another.
- **Auxiliary Problem:** A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.

2 Problem Types

- **Finding an Initial Feasible Solution:** This problem addresses situations where the origin is not a feasible solution to the linear programming problem, requiring a method to find a starting feasible point for algorithms like the Simplex method.
- **Handling Infeasible Linear Programs:** This problem involves determining whether a given linear program has any feasible solutions at all, and if not, providing a method to determine this.
- **Solving Linear Programs Graphically in Two Dimensions:** This problem focuses on solving linear programming problems with two variables by visualizing the feasible region and identifying optimal solutions at vertices.
- **Solving Linear Programs Using the Simplex Method:** This problem involves applying the Simplex algorithm to iteratively find an optimal solution by moving between vertices of the feasible region, updating a dictionary of variables in each iteration.
- **Understanding the Simplex Dictionary:** This problem involves understanding the structure and interpretation of the Simplex dictionary, which represents the basic and non-basic variables and their relationships in a linear program.

3 Algorithm Solutions

- **Simplex Method:** Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This corresponds to moving from vertex to vertex in the feasible region.
- **Auxiliary Problem:** Introduce a new variable x_0 and modify constraints to create a new problem where the origin is feasible. Solve this auxiliary problem to find a feasible solution for the original problem. If $x_0 = 0$ in the optimal solution of the auxiliary problem, then the solution is feasible for the original problem.